

SPENCER DELCORE

Computer Engineering - University of Waterloo

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QUALIFICATIONS

Languages	C++, Python, C, C#, VHDL/Verilog, MATLAB, PHP, JavaScript
Frameworks/Technologies	Qt, QML, Git, Unity, AWS, SQL, Xilinx PetaLinux
Embedded Experience	FPGAs, UNIX, Arduino, Raspberry Pi, Avnet MiniZed

WORK EXPERIENCE

Evertz Microsystems Ltd. - *Embedded Systems Design Engineering, Co-op* Sept. 2017 - Dec. 2017
Burlington, ON

- Developed a multi-threaded framework for generating embedded user interfaces using Qt and QML
- Implemented an embedded SQLite database, and configured it for performance
- Communicated through a defined protocol with various systems via D-Bus
- Parsed multiple JSON structures to generate various user interfaces

Cognitive Systems Corp. - *Augmented Reality (AR) Developer, Co-op* Jan. 2017 - April. 2017
Quantum Valley Investments, Waterloo, ON

- Responsible for creating an AR demonstration to visualize wireless signals using a Microsoft HoloLens
- Dynamically generated interactive holograms using Unity and C#
- Created a Python program that interfaces with Cognitive's system on a chip (SoC) to detect wireless signals
- Implemented websockets to communicate between the SoC and the HoloLens

Ingle International Inc. - *Software Engineering, Co-op* May. 2016 - Aug. 2016
Toronto, ON

- Accountable for building and maintaining version two of their API using PHP and the Laravel framework
- Built a dynamically themed Android application using Java and React Native
- Researched, presented, and participated in critical development meetings

VIZIYA - *Software Developer, Co-op* Sept. 2015 - Dec. 2015
Waterdown, ON

- Designed and developed quality ASP.NET and C# applications
- Performed SQL queries to retrieve and manipulate data from Oracle and SAP databases
- Generated user guides, training manuals, and documentation

Emerging Star Investment Group - *Software Developer, Co-op* Jan. 2015 - April 2015
Accelerator Centre, Waterloo, ON

- Developed financial web-based software using PHP and the Laravel framework
- Created and maintained an Amazon Web Services (AWS) EC2 server to host the company website
- Built scripts to scrape and store financial related data using Python

PROJECTS

VHDL Compiler - *Java, VHDL* 2017

- Created a VHDL synthesizer and simulator using Java
- Uses VHDL (combinational circuits only) as the input language and produces circuit diagrams and simulations
- Gathered important compiler concepts such as regular languages, regex, DFA, lexing, recursive decent parsing, and context-free grammars

DAC Audio Player - *Altera FPGA, C* 2016

- Implemented a 16-bit sampling stereo digital to analog audio player with push buttons to control playback
- Gained experience with Altera audio IP core, audio sampling and buffering, and interrupts
- Made use of the FatFS CLI to mount and interpret WAV files from a microSD card